

MODULE 4A • EPISODE 1

Gravity and the Solar System

From Falling Bodies to Kepler's Ellipses

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Gravity and the Solar System

This episode follows one force from a dropped object to the layout of the Solar System. It begins with Galileo's Law of Falling Bodies and asks why every freely falling object near Earth's surface speeds up at the same steady rate. Newton supplies the answer: his Second Law of Motion and his Law of Universal Gravitation together show that the acceleration due to gravity, g , depends on Earth's mass and radius but not on the mass of the falling object. The panel then follows gravity upward, to the International Space Station and the Moon, uses Newton's Third Law to explain the tides, and shows why g is slightly smaller at the equator than at the poles. The episode closes with the 16th- and 17th-century debate between the geocentric and heliocentric models, Kepler's discovery that orbits are ellipses, and the flat, disk-shaped arrangement of the Solar System that places the planets in the Zodiac.

The Panel

Name	Role
Dr. Edward Thorne	Host; astrophysicist
Dr. Lena Petrova	Guest; exoplanet researcher
Dr. Kenji Tanaka	Guest; microbial biologist with a strong interest in astrobiology

Learning Objectives

After this episode you should be able to:

- ◆ State Galileo's Law of Falling Bodies and describe uniform acceleration near Earth's surface.
- ◆ State Newton's Second Law and use $W = mg$ to relate weight, mass, and the acceleration due to gravity.
- ◆ State Newton's Law of Universal Gravitation and explain how it leads to $g = GM/R^2$.
- ◆ Explain why g does not depend on the mass of a falling object, and how this explains Galileo's Law.
- ◆ Describe how g changes with altitude, and explain why astronauts on the Space Station feel weightless.
- ◆ Apply Newton's Third Law to the Earth and Moon, and explain how the Moon produces tides, including spring and neap tides.
- ◆ Explain the two reasons g is smaller at the equator than at the poles.
- ◆ Compare the geocentric and heliocentric models and describe the contributions of Copernicus and Kepler.
- ◆ Define the visible planets, the Ecliptic Plane, and the Zodiac, and name other objects that orbit the Sun.

QUICK REFERENCE

Key Terms

Term	Definition
Galileo's Law of Falling Bodies	Excluding aerodynamic factors, objects of different mass released from the same height hit the ground at the same time.
Freely falling object	An object falling only under the influence of gravity.
Uniform acceleration	Acceleration that stays constant; speed changes by the same amount every second.
Acceleration due to gravity (g)	The acceleration of a freely falling object, about 9.8 m/s^2 near Earth's surface.
Newton's Second Law of Motion	When a net external force acts on an object, the object accelerates: $F = ma$.
Net force	The total of all the external forces acting on an object.
Weight (W)	The gravitational force on an object; $W = mg$.
Newton's Law of Universal Gravitation	Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centers.
Universal gravitational constant (G)	The constant in Newton's Law of Universal Gravitation. Not the same as g.
Free fall	Motion under gravity alone. Astronauts in orbit are in free fall, which is why they feel weightless.
Newton's Third Law of Motion	For every action, there is an equal and opposite reaction.
Tides	The rise and fall of the oceans, caused mainly by the Moon pulling harder on Earth's near side than on its far side.
Spring tides	The largest tidal range, when the Sun, Earth, and Moon are aligned (new and full moon).
Neap tides	The smallest tidal range, when the Sun and Moon are at right angles as seen from Earth (quarter moons).
Effective weight	The weight an object seems to have once the effect of Earth's rotation is included.
Equatorial bulge	Earth's slight flattening; the equator is about 21 km farther from Earth's center than the poles are.
Geocentric model	The model of Ptolemy, with the Earth at the center.
Heliocentric model	The model with the Sun at the center, revived by Copernicus in 1543.
Epicycles	Small extra circles used to make circular-orbit models match the observed motions of the planets.

Term	Definition
Ellipse	An oval curve with two foci. Kepler showed that each planet's orbit is an ellipse with the Sun at one focus.
Visible planets	Planets that can be seen with the naked eye: Mercury, Venus, Mars, Jupiter, and Saturn, plus Earth.
Ecliptic Plane	The plane of Earth's orbit around the Sun, used as the reference plane of the Solar System.
Zodiac	The band of sky about eight degrees north and south of the Ecliptic Plane, through which the Sun, Moon, and planets appear to move.
Other Solar System objects	Asteroids, comets, Kuiper Belt Objects, and the icy bodies of the Oort Cloud, all of which orbit the Sun.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
9.8 m/s²	Acceleration due to gravity near Earth's surface
9.8, 19.6 m/s	Speed of a dropped object after 1 and 2 seconds
~250 mi (~400 km)	Altitude of the International Space Station
~8.7 m/s²	Value of g at the Space Station, about 11% less than at the surface
~2 × 10²⁰ N	Earth's gravitational force on the Moon (and the Moon's on Earth)
9.832 m/s²	Value of g at the poles
9.780 m/s²	Value of g at the equator
9.803 m/s²	Value of g at mid-latitudes such as Chicago
~21 km	How much farther the equator is from Earth's center than the poles are
1543	Publication of Copernicus's On the Revolutions of the Heavenly Spheres
~8°	Width of the Zodiac north and south of the Ecliptic Plane
5 + Earth	The visible planets; Uranus and Neptune generally need a telescope

EPISODE CONTENT

Part-by-Part Summary

Part 1 — Gravity and Falling Bodies

Gravity is the force that shapes the entire Universe, from the orbits of the planets to the structure of galaxies. The episode starts close to home, with **Galileo's Law of Falling Bodies**: excluding aerodynamic factors, objects of different mass released from the same height hit the ground at the same time.

The reason is that **freely falling objects**, those falling only under the influence of gravity, fall with a **constant, or uniform, acceleration** of about **9.8 meters per second, every second**. Starting from rest, a falling object moves at 9.8 m/s after one second, 19.6 m/s after two seconds, and so on.

Part 2 — Newton's Laws of Motion and Gravity

Newton's Second Law of Motion says that when a net external force acts on an object, the object accelerates: $F = ma$. For a freely falling object, the only force is gravity, which we call its **weight**, W , so the Second Law becomes $W = mg$, where g is the acceleration due to gravity.

Weight itself comes from **Newton's Law of Universal Gravitation**: every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centers. Your weight is the gravitational force between your mass and Earth's.

Combining the two laws gives $g = GM/R^2$, where G is the universal gravitational constant, M is Earth's mass, and R is Earth's radius. The falling object's own mass cancels out, and M and R are essentially constant near the surface, so every freely falling body there has the same constant acceleration, whatever its mass. This **explains Galileo's Law**.

Part 3 — Gravity Beyond the Surface

Because gravity weakens with distance, g declines as we move away from Earth. At the **International Space Station**, about 250 miles up, g is about **8.7 m/s²**, only about 11% less than at the surface. The astronauts still have plenty of weight; they feel weightless because they and the station are in a constant state of **free fall**.

Even at the distance of the Moon, g isn't zero. Earth pulls on the Moon with a force of about **2×10^{20} newtons**, the force that keeps the Moon in orbit.

Newton's Third Law of Motion says that for every action there is an equal and opposite reaction, so the Moon pulls on Earth with the same force. Because the Moon pulls a little harder on the side of Earth facing it than on the far side, the pull stretches the oceans slightly, producing the **tides**.

The Sun contributes too. When the Sun, Earth, and Moon are aligned, at new moon and full moon, we get the largest tides, the **spring tides**. When the Sun and Moon are at right angles, at the quarter moons, we get the smaller **neap tides**.

Part 4 — Gravity and Latitude

The measured value of g also changes with latitude, mostly because Earth rotates. Every object away from the poles travels in a circle once a day, and moving in a circle takes a small inward force. A little of Earth's gravitational pull is used up providing it, leaving less to press the object against the ground. Objects at the equator make the biggest circles and have the greatest reduction in **effective weight**; objects at the poles have none.

A second effect adds to the first: Earth **bulges slightly at the equator**, so someone standing there is about 21 kilometers farther from Earth's center than someone at a pole, which lowers g a little more.

The differences are small but measurable: g is about **9.832 m/s²** at the poles, **9.780 m/s²** at the equator, and about **9.803 m/s²** at mid-latitudes such as Chicago. Your weight changes slightly depending on where you are on Earth.

Part 5 — The Solar System Models

In the 16th and 17th centuries, astronomers argued over two models: the **geocentric model** of Ptolemy, with the Earth at the center, and the **heliocentric model**, with the Sun at the center. The Sun-centered idea goes back to the ancient Greek astronomer **Aristarchus**, but it was the Polish cleric **Copernicus** who revived it as a complete Solar System model in *On the Revolutions of the Heavenly Spheres*, published in 1543 as he lay dying.

Copernicus was right to put the Sun at the center, but he assumed the planets moved in perfect circles. To match the sky he still needed **epicycles**, the small extra circles Ptolemy had used, so in practice his model was not much simpler or more accurate than Ptolemy's, which slowed its acceptance.

Johannes Kepler, a contemporary of Galileo, fixed the flaw by showing that planetary orbits are **ellipses** with the Sun at one focus. That insight, together with Galileo's telescope observations, led to the general acceptance of the heliocentric model.

The **visible planets** are those seen with the naked eye: Mercury, Venus, Mars, Jupiter, and Saturn, plus Earth itself. Uranus and Neptune generally require a telescope; Uranus is at the edge of naked-eye visibility under very dark skies.

Because the Solar System formed from a thin disk of gas and dust, the planets orbit in roughly the same plane. The plane of Earth's orbit serves as the reference, the **Ecliptic Plane**, and the band about eight degrees north and south of it is the **Zodiac**. The planets' orbits lie within it, which is why the Sun, Moon, and planets move through the zodiacal constellations. Asteroids, comets, Kuiper Belt Objects, and the icy bodies of the Oort Cloud also orbit the Sun.

Part 6 — Conclusion

The episode traces one idea from Galileo's falling bodies to Newton's universal laws, showing that the same gravity that pulls a dropped object to the ground also holds the Moon in orbit and raises the tides. It sets the stage for a closer look at Kepler's revolutionary insights into how the planets move.

STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Part 1 — Gravity and Falling Bodies

1. State Galileo's Law of Falling Bodies.

ANSWER Excluding aerodynamic factors such as air resistance, objects of different mass released from the same height hit the ground at the same time.

2. What is a freely falling object, and what kind of acceleration does it have?

ANSWER An object falling only under the influence of gravity. It has a constant, or uniform, acceleration.

3. What does it mean to say that g is “9.8 meters per second, every second”?

ANSWER A falling object's speed increases by 9.8 m/s during each second it falls. The units are written m/s².

4. An object is dropped from rest. Ignoring air resistance, how fast is it falling after 1, 2, and 3 seconds?

ANSWER 9.8 m/s, 19.6 m/s, and 29.4 m/s.

Part 2 — Newton's Laws of Motion and Gravity

1. State Newton's Second Law of Motion and identify each symbol in $F = ma$.

ANSWER When a net external force acts on an object, the object accelerates. F is the net force, m is the object's mass, and a is its acceleration.

2. What is weight? Write the equation that relates weight to mass.

ANSWER Weight is the gravitational force on an object, the only force acting on a freely falling object. $W = mg$, where g is the acceleration due to gravity.

3. State Newton's Law of Universal Gravitation.

ANSWER There is a universally attractive force between any two masses. Its magnitude is proportional to the product of the two masses and inversely proportional to the square of the distance between their centers.

4. If the distance between the centers of two masses doubles, what happens to the gravitational force between them?

ANSWER It becomes one-fourth as large, because the force depends on $1/\text{distance}^2$ and $2^2 = 4$.

5. Which two laws are combined to find g , and what expression results?

ANSWER Newton's Second Law (in the form $W = mg$) and the Law of Universal Gravitation. Together they give $g = GM/R^2$, where G is the universal gravitational constant, M is Earth's mass, and R is Earth's radius.

6. Explain why g does not depend on the mass of the falling object, and how this explains Galileo's Law.

ANSWER The object's mass appears in both $W = mg$ and the gravitation law, so it cancels when they are combined. What remains depends only on Earth's mass and radius, which are essentially constant near the surface. So every freely falling object there has the same constant acceleration and, released together, they land together.

Part 3 — Gravity Beyond the Surface

1. What happens to g as you move far above Earth's surface? Why?

ANSWER It decreases, because the distance from Earth's center increases and gravity weakens with the square of that distance.

2. What is g at the International Space Station, about 250 miles up, and how does it compare with the surface value?

ANSWER About 8.7 m/s^2 , roughly 11% less than at the surface.

3. If gravity at the Space Station is nearly 90% of its surface value, why do the astronauts feel weightless?

ANSWER They still have weight, but they and the station are in a constant state of free fall, falling together around Earth, so nothing presses against them.

4. Is g zero at the distance of the Moon? How large is Earth's gravitational force on the Moon, and what does it do?

ANSWER No. Earth pulls on the Moon with about 2×10^{20} newtons, the force that keeps the Moon in orbit around Earth.

5. State Newton's Third Law of Motion and apply it to the Earth and Moon.

ANSWER For every action there is an equal and opposite reaction. If Earth pulls on the Moon with about $2 \times 10^{20} \text{ N}$, the Moon pulls on Earth with an equal force in the opposite direction.

6. Explain how the Moon's gravity produces the tides.

ANSWER The Moon pulls a little harder on the side of Earth facing it than on Earth's far side. That difference stretches the oceans slightly, producing the tides.

7. What are spring tides and neap tides, and when does each occur?

ANSWER Spring tides are the largest tides, with the highest highs and lowest lows; they occur when the Sun, Earth, and Moon are aligned, at new moon and full moon. Neap tides are the smallest; they occur when the Sun and Moon are at right angles as seen from Earth, at the quarter moons.

Part 4 — Gravity and Latitude

1. Why does Earth's rotation reduce the measured value of g ?

ANSWER Objects on the rotating Earth travel in a circle once a day, which takes a small inward force. A little of Earth's gravitational pull is used up providing that force, leaving less to press the object against the ground.

2. Where on Earth is the reduction from rotation greatest, and where is it zero? Why?

ANSWER Greatest at the equator, where objects travel in the biggest circles. Zero at the poles, where objects make no circles.

3. What second effect makes g smaller at the equator than at the poles?

ANSWER Earth bulges slightly at the equator, so a person there is about 21 kilometers farther from Earth's center, and gravity is a little weaker.

4. Give the approximate value of g at the poles, at the equator, and at mid-latitudes such as Chicago.

ANSWER About 9.832 m/s^2 at the poles, 9.780 m/s^2 at the equator, and 9.803 m/s^2 at mid-latitudes.

5. About what percentage less would you weigh at the equator than at the North Pole?

ANSWER $(9.832 - 9.780) / 9.832 \approx 0.005$, or about half a percent.

Part 5 — The Solar System Models

1. Distinguish the geocentric and heliocentric models. Which astronomer is associated with the geocentric model?

ANSWER The geocentric model places the Earth at the center; the heliocentric model places the Sun at the center. The geocentric model is associated with Ptolemy.

2. Who first proposed a Sun-centered system, and who revived it as a complete Solar System model? When, and in what book?

ANSWER The ancient Greek astronomer Aristarchus first proposed it. Copernicus revived it in *On the Revolutions of the Heavenly Spheres*, published in 1543.

3. What was the flaw in Copernicus's model, and what was its consequence?

ANSWER He assumed that planetary orbits were perfect circles. To match the sky he still needed epicycles, so his model was not much simpler or more accurate than Ptolemy's, which slowed its acceptance.

4. Who fixed the flaw, and how? What else helped the heliocentric model win acceptance?

ANSWER Johannes Kepler, a contemporary of Galileo, showed that planetary orbits are ellipses with the Sun at one focus. Galileo's telescope observations also supported the heliocentric model.

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5. What are the visible planets? Which planets generally require a telescope?

ANSWER Mercury, Venus, Mars, Jupiter, and Saturn, which can be seen with the naked eye, plus Earth. Uranus and Neptune generally require a telescope, although Uranus is at the edge of naked-eye visibility under very dark skies.

6. Why do the planets orbit in roughly the same plane? What is the Ecliptic Plane?

ANSWER The Solar System formed from a relatively thin disk of gas and dust. The Ecliptic Plane is the plane of Earth's orbit around the Sun, used as the reference plane.

7. What is the Zodiac, and why do the Sun, Moon, and planets appear in the zodiacal constellations?

ANSWER The band of sky about eight degrees north and south of the Ecliptic Plane. The planets' orbits lie within it, so the Sun, Moon, and planets all appear to move through its constellations.

8. Besides the planets, name four kinds of objects that orbit the Sun.

ANSWER Asteroids, comets, Kuiper Belt Objects, and the icy bodies of the Oort Cloud.

Part 6 — Conclusion

1. The episode says the same gravity that pulls a dropped object to the ground also holds the Moon in orbit. Explain how the episode supports this idea.

ANSWER Universal gravitation applies to any two masses. It gives the constant g that makes objects fall together near the surface; it weakens with distance but is not zero at the Moon, where it supplies the force that keeps the Moon in orbit; and by the Third Law the Moon pulls back on Earth, raising the tides. One law covers falling bodies, orbits, and tides.

2. Kepler's ellipses came decades before Newton's laws. Why does the episode treat Newton and Kepler as parts of one story?

ANSWER Kepler found the correct shape of planetary orbits, which made the heliocentric model work. Newton's Law of Universal Gravitation later explained why orbits have that shape: the same force that governs falling bodies governs the planets.

TEACHER NOTES

Corrections and Cautions

Points in the episode that need a qualification, or a prepared answer for the question a sharp student will ask.

C1 Little g and big G.

Students often confuse g, the acceleration due to gravity (9.8 m/s^2 near the surface), with G, the universal gravitational constant, $6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$. The episode never gives G's value. g changes from place to place; G does not.

C2 Units of g.

"Meters per second, every second" and "meters per second squared" are the same thing. The episode uses both, so point out the equivalence before students see m/s^2 in Parts 3 and 4.

C3 $g = GM/R^2$ is a near-surface formula.

Far above the surface, R must be replaced by the distance from Earth's center, $R + h$. That is how the Space Station value is found: $9.8 \times (6371/6771)^2 \approx 8.7 \text{ m/s}^2$. The Moon, about 60 Earth radii away, feels $g \approx 9.8/60^2 \approx 0.0027 \text{ m/s}^2$ — small, but not zero.

C4 "Weightless" astronauts.

Students may have heard that there is no gravity in space. The episode is careful to say the astronauts still have weight. If asked why the station doesn't fall to Earth: it does, continually, but it moves sideways fast enough (about 17,500 mph) that it keeps missing.

C5 Why tides come in pairs.

Because tides come from the difference in the Moon's pull across Earth, the oceans bulge on both the near side and the far side. Most coasts get two high tides a day, about 12 hours 25 minutes apart. The Third Law explains why the Moon pulls on Earth at all; the tides come from how that pull varies across Earth's diameter.

C6 Spring tides have nothing to do with spring.

The name refers to the water "springing up." They happen twice a month, at new and full moon. The Sun's tidal effect is a little less than half the Moon's, even though the Sun's total pull on Earth is far larger.

C7 Two causes of the latitude effect.

Rotation accounts for roughly two-thirds of the difference between the poles and the equator ($\omega^2 R \approx 0.034 \text{ m/s}^2$ at the equator); the equatorial bulge accounts for the rest. Local geology and altitude cause further small variations, so the Chicago figure is a typical value, not an exact one.

C8 Aristarchus and Copernicus.

Aristarchus of Samos proposed a Sun-centered system around 250 BCE, but it found few followers. Copernicus's model kept epicycles and predicted planetary positions about as well as Ptolemy's; its advantages were explanatory, such as a natural account of retrograde motion.

C9 Kepler's ellipses are only slightly oval.

Most planetary orbits are close to circles; Earth's eccentricity is 0.017. Kepler could detect the difference for Mars (eccentricity 0.093) only because of Tycho Brahe's unusually precise naked-eye observations. Textbook diagrams exaggerate the ovals.

C10 Earth as a visible planet.

The classical naked-eye planets are Mercury, Venus, Mars, Jupiter, and Saturn. The episode adds Earth to complete the list of six. Uranus, at about magnitude 5.7, is just visible to sharp eyes under very dark skies; Neptune never is.

C11 The Zodiac's constellations.

The Sun's path actually crosses 13 constellations, including Ophiuchus, and the astronomical boundaries differ from the 12 equal astrological signs. The eight-degree width is traditional; Venus and Mercury can occasionally stray to its edge.

C12 The Ecliptic Plane is a reference, not the exact middle.

The average plane of all the planets' orbits, the invariable plane, is tilted about 1.6° from the ecliptic. Astronomers use Earth's orbit as the reference simply because we observe from Earth.

TEACHER NOTES

Teaching the Episode**Start with the drop.**

Run the book-and-paper demonstration from the Try It box, then show the Apollo 15 video in which astronaut David Scott drops a hammer and a feather on the Moon in 1971. With no air, they land together.

Let the mass cancel on the board.

Write $mg = GMm/R^2$ and cross out the m on each side. Students who see the cancellation happen usually remember why Galileo was right.

Compute g at the Space Station together.

Using Earth's radius, 6,371 km, and an altitude of 400 km, have students find the ratio $(6371/6771)^2 \approx 0.885$. Then ask them to explain, in one sentence, why astronauts float anyway.

Newton's Moon test.

The Moon is about 60 Earth radii away, so gravity there should be $1/3600$ of its surface value. Newton checked that this weakened pull matches the acceleration needed to keep the Moon in its orbit — the same check this episode makes in words.

Weigh yourself around the world.

Ask how much less a 150-pound person would weigh at the equator than at the North Pole (about 0.8 pound). Then ask why a bathroom scale would never show it.

Find the ecliptic in the sky.

On a night when two or more planets are visible, have students sketch them with the Moon. The objects fall along a line across the sky, the ecliptic seen edge-on.

Use the panel.

Kenji brings an astrobiologist's angle: a planet's gravity helps decide whether it can hold an atmosphere and liquid water. Ask students why a small world like the Moon has almost no air.